

Design and Evaluation of a Soft Exoskeleton Finger with Hybrid Jamming towards Hand Rehabilitation

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Context

- Everyone in this room is getting older.
- 1 in 4 people above the age of 25 will experience a stroke in their lifetime. (Feigin et al., 2025)
 - Less than 20% will regain full upper limb function. (Graham, 2024)
- We all rely heavily on our hands for daily activities (Graham, 2024)



Context



Credit: Wyss Institute at Harvard University

Soft robotic exoskeleton gloves have emerged as a promising rehab technology

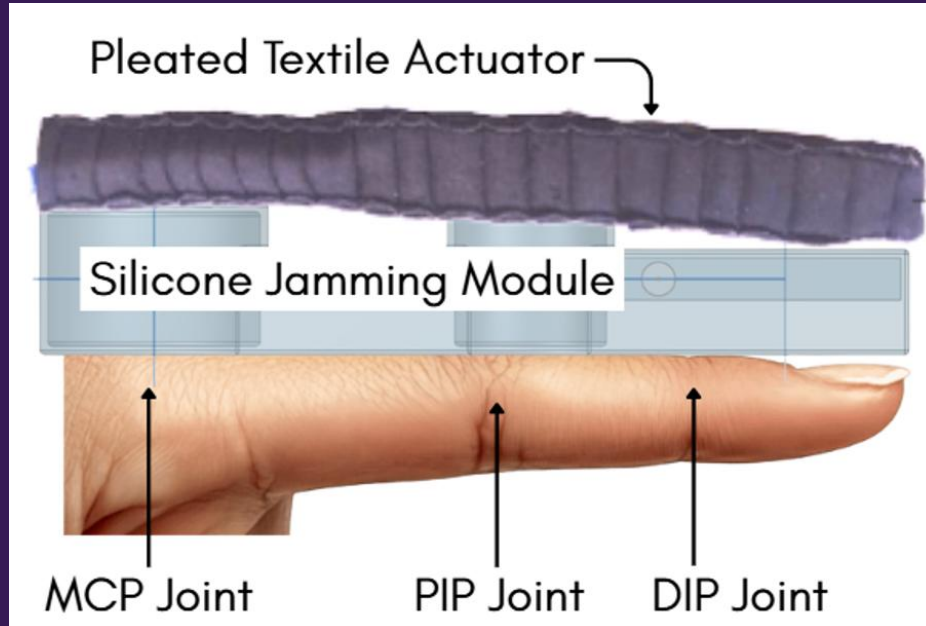


Credit: iRobot

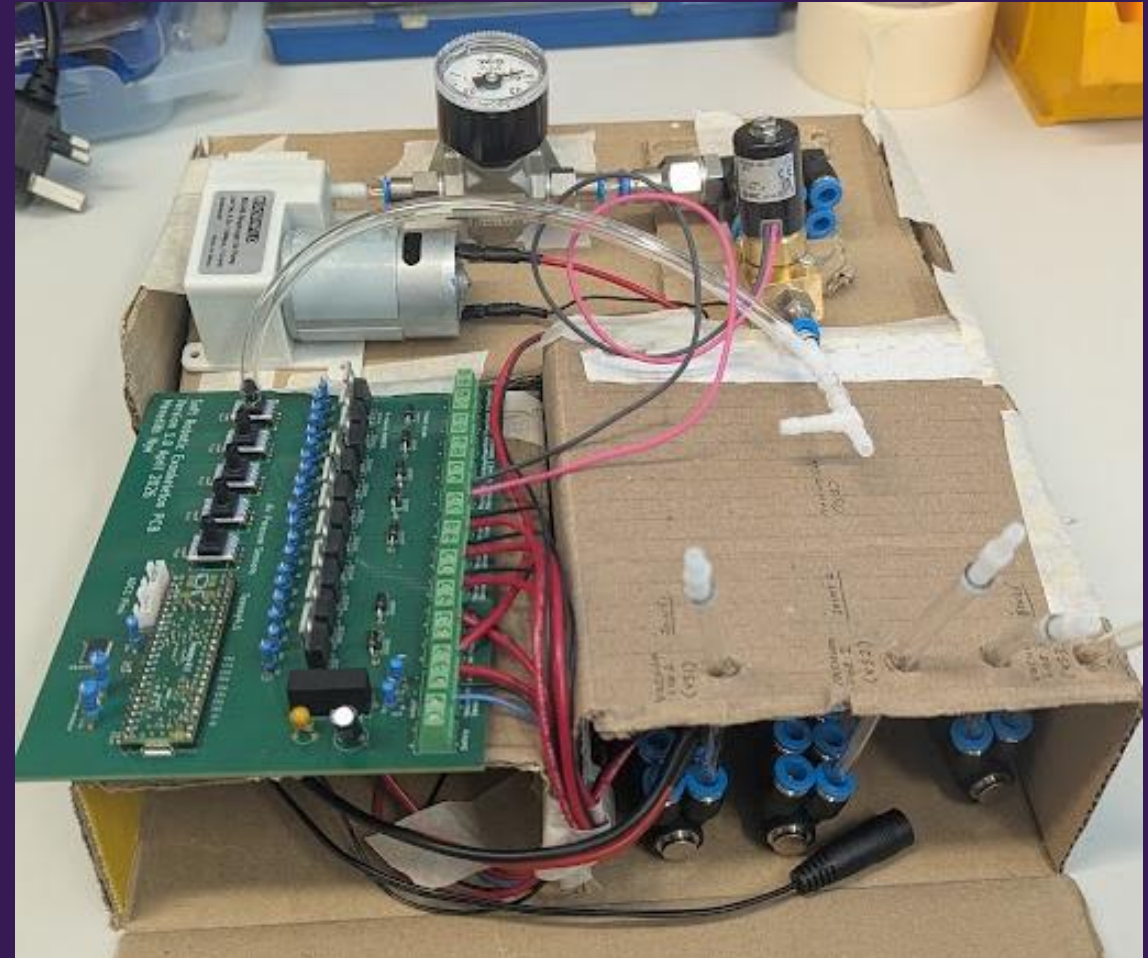
Jamming is a novel technique that allows forms to go from soft to rigid with changes in air pressure.

My design

Finger level design



Electronics and Pneumatics System



Design Aims- Lab based project

Investigate a soft exoskeleton finger design using granular and laminar jamming.

- Can hybrid jamming provide new exoskeleton characteristics?
- How does the finger meet rehabilitation requirements?
- How does the finger compare to other designs?

Manufacturing and Methods

Prototype testing:

- Measuring force output at the finger's tip
- Measuring stiffness modulation
- Actuation speed
- Power consumption



Outcomes

Expected results:

- Joint level range of motion achievement while different jamming pockets are activated
- Force output within the 8 - 9N range
- 5 -10x stiffness variation

Takeaways so far:

- The manufacturing process is a bottleneck for the finger design.
- Novel system complexity adds cost, weight, and manufacturing time.